

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 4

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1. TITLE

Develop Dynamic Webpage Using Java Script

2. AIM

- To develop a dynamic webpage using **HTML, CSS, and JavaScript**.
 - To create a **To-Do List application** that allows users to add, complete, and delete tasks dynamically.
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3. OBJECTIVE

The objectives of this experiment are:

- To understand the concept of developing dynamic webpages using JavaScript.
 - To use JavaScript functions to perform different operations.
 - To dynamically create and add HTML elements using JavaScript.
 - To implement functionality for adding tasks to a To-Do List.
 - To mark completed tasks using JavaScript and CSS classes.
 - To delete tasks dynamically from the webpage.
 - To use DOM methods to access, create, modify, and remove HTML elements.
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4. THEORY

JavaScript is a programming language used to make webpages dynamic and interactive. It allows webpage content to be created, modified, and removed based on user actions.

In this application, JavaScript is used to develop a dynamic **To-Do List** that allows users to manage their daily tasks.

- **HTML** is used to create the structure of the To-Do List.
- **CSS** is used to style the webpage, tasks, and buttons.
- **JavaScript functions** are used to perform task-related operations.
- **DOM methods** are used to access and modify HTML elements dynamically.
 - `document.createElement()` is used to create new task elements.
 - `appendChild()` is used to add tasks and buttons to the webpage.
 - `classList.toggle()` is used to mark a task as completed or uncompleted.
 - `remove()` is used to delete a task from the list.
- **Event handling** is used to perform actions when buttons are clicked.

The basic flow of the application is:

**User Input → Add Task Function → Dynamic HTML Element Creation → Display Task
→ Complete/Delete Action → Updated To-Do List**

5. ALGORITHM / PROCEDURE

1. Create a new HTML file.
 2. Add the basic HTML document structure using `<!DOCTYPE html>`, `<html>`, `<head>`, and `<body>`.
 3. Add CSS styles to design the To-Do List webpage.
 4. Create a container to hold the To-Do List application.
 5. Add an input field to accept tasks from the user.
 6. Add an **Add** button to call the `addTask()` function.
 7. Create an empty unordered list to display the tasks.
 8. Use `document.getElementById()` to access the task input field.
 9. Check whether the entered task is empty.
 10. Display an alert message if no task is entered.
 11. Call the `createTask()` function when a valid task is entered.
 12. Use `document.createElement()` to dynamically create a list item, task text, Complete button, and Delete button.
 13. Use `appendChild()` to add the task and buttons to the list item.
 14. Add the newly created task to the To-Do List.
 15. Use `classList.toggle()` to mark a task as completed or uncompleted.
 16. Use `remove()` to delete a task from the list.
 17. Clear the input field after adding a task.
 18. Open the HTML file in a web browser.
 19. Add, complete, and delete different tasks to verify the dynamic functionality.
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6. PROGRAM

Main.html:

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
initial-scale=1.0">
  <title>F1 2026 Team Explorer</title>

  <link rel="stylesheet" href="style.css">
</head>

<body>

  <h1>Formula 1 Team Explorer 2026</h1>

  <p>Enter or Select an F1 Team:</p>

  <input
    type="text"
    id="teamInput"
    placeholder="Type team name">

  <p>OR</p>

  <select id="teamSelect" onchange="clearInput()">
    <option value="">-- Select Team --</option>
    <option value="mclaren">McLaren</option>
    <option value="ferrari">Ferrari</option>
    <option value="mercedes">Mercedes</option>
    <option value="red bull">Red Bull</option>
    <option value="williams">Williams</option>
    <option value="aston martin">Aston Martin</option>
    <option value="alpine">Alpine</option>
```

```
<option value="haas">Haas</option>  
<option value="racing bulls">Racing Bulls</option>  
<option value="Audi">Audi</option>
```

```
    <option value="cadillac">Cadillac</option>  
</select>
```

```
<br><br>

<button onclick="showTeam() ">
    Show Details
</button>

<hr>

<div id="card">

    <h2>Team Name</h2>

    <p>Drivers:</p>

    <p>Engine:</p>

    <p>Position:</p>

    <p>Wins:</p>

    <p>Points:</p>

</div>

<script src="main.js"></script>

</body>

</html>
```

Script js :

```
function clearInput() {
    document.getElementById("teamInput").value = "";
}
function showTeam() {

    let team = document.getElementById("teamInput").value;

    if (team == "") {
        team = document.getElementById("teamSelect").value;
    }

    team = team.toLowerCase();

    if (team == "mclaren") {

        document.getElementById("card").innerHTML = `
            <h2>McLaren</h2>
            <p><b>Drivers:</b> Lando Norris, Oscar Piastri</p>
            <p><b>Engine:</b> Mercedes</p>
            <p><b>Position:</b> 3rd</p>
            <p><b>Wins:</b> 1</p>
            <p><b>Points:</b> 220 points</p>
        `;

    }

    else if (team == "ferrari") {

        document.getElementById("card").innerHTML = `
            <h2>Ferrari</h2>
            <p><b>Drivers:</b> Charles Leclerc, Lewis Hamilton</p>
            <p><b>Engine:</b> Ferrari</p>
            <p><b>Position:</b> 2nd</p>
            <p><b>Wins:</b> 2</p>
            <p><b>Points:</b> 307 points</p>
        `;

    }

}
```

```
}  
else if (team == "mercedes") {
```

```
document.getElementById("card").innerHTML = `
    <h2>Mercedes</h2>
    <p><b>Drivers:</b> George Russell, Kimi Antonelli</p>
    <p><b>Engine:</b> Mercedes</p>
    <p><b>Position:</b> 1st</p>
    <p><b>Wins:</b> 8</p>
    <p><b>Points:</b> 379 points</p>
`;

}

else if (team == "red bull") {

    document.getElementById("card").innerHTML = `
        <h2>Red Bull</h2>
        <p><b>Drivers:</b> Max Verstappen, Isack Hadjar</p>
        <p><b>Engine:</b> Red Bull Ford Powertrains</p>
        <p><b>Position:</b> 4th</p>
        <p><b>Wins:</b> 0</p>
        <p><b>Points:</b> 177 points</p>
    `;

}

else if (team == "williams") {

    document.getElementById("card").innerHTML = `
<h2>Williams</h2>

        <p><b>Drivers:</b> Alex Albon, Carlos Sainz</p>
        <p><b>Engine:</b> Mercedes</p>
        <p><b>Position:</b> 9th</p>
        <p><b>Wins:</b> 0</p>
        <p><b>Points:</b> 11 points</p>
    `;

}

else if (team == "aston martin") {

    document.getElementById("card").innerHTML = `
        <h2>Aston Martin</h2>
    `;
}
```

<p>Drivers: Fernando Alonso, Lance Stroll</p>

<p>Engine: Honda</p>

<p>Position: 10th</p>

<p>Wins: 0</p>


```
        <p><b>Points:</b> 1 point</p>
    `;

}

else if (team == "alpine") {

    document.getElementById("card").innerHTML = `
        <h2>Alpine</h2>
        <p><b>Drivers:</b> Pierre Gasly, Franco Colapinto</p>
        <p><b>Engine:</b> Mercedes</p>
        <p><b>Position:</b> 6th</p>
        <p><b>Wins:</b> 0</p>
        <p><b>Points:</b> 61 points</p>
    `;

}

else if (team == "haas") {

    document.getElementById("card").innerHTML = `
        <h2>Haas</h2>
        <p><b>Drivers:</b> Esteban Ocon, Oliver Bearman</p>
        <p><b>Engine:</b> Ferrari</p>
        <p><b>Position:</b> 7th</p>
        <p><b>Wins:</b> 0</p>
        <p><b>Points:</b> 21 points</p>
    `;

}

else if (team == "racing bulls") {

    document.getElementById("card").innerHTML = `
        <h2>Racing Bulls</h2>
        <p><b>Drivers:</b> Arvid Lindblad, Liam Lawson</p>
        <p><b>Engine:</b> Red Bull Ford Powertrains</p>
        <p><b>Position:</b> 5th</p>
        <p><b>Wins:</b> 0</p>
        <p><b>Points:</b> 66 points</p>
    `;

}
```

```
}  
else if (team == "audi") {
```

```
document.getElementById("card").innerHTML = `
    <h2>Audi</h2>
    <p><b>Drivers:</b> Nico Hülkenberg, Gabriel Bortoleto</p>
    <p><b>Engine:</b> Audi</p>
    <p><b>Position:</b> 8th</p>
    <p><b>Wins:</b> 0</p>
    <p><b>Points:</b> 12 points</p>
`;

}

else if (team == "cadillac") {

    document.getElementById("card").innerHTML = `
        <h2>Cadillac</h2>
        <p><b>Drivers:</b>Sergio Pérez, Valtteri Bottas</p>
        <p><b>Engine:</b>Ferrari</p>
        <p><b>Position:</b> 22nd</p>
        <p><b>Wins:</b> 0</p>
        <p><b>Points:</b> 0</p>
    `;

}

else {

    document.getElementById("card").innerHTML = `
<h2>Team Not Found</h2>

        <p>Please enter a valid Formula 1 team.</p>
    `;

}

}
```

Style.css:

```
/* Page */

body{
    font-family: Arial, Helvetica, sans-serif;
    background-color: #f4f4f4;
    text-align: center;
    margin: 0;
    padding: 30px;
}

/* Heading */

h1{
    color: #e10600;
}

/* Input and Dropdown */

input, select{
    width: 250px;
    padding: 10px;
    margin: 10px;
    font-size: 16px;
    border-radius: 8px;
border: 2px solid #ccc;
}

/* Button */

button{
    background-color: #e10600;
    color: white;

    border: none;
    padding: 12px 25px;
    font-size: 16px;
    border-radius: 8px;
    cursor: pointer;
}
```

```
}
```

```
button:hover{
```

```
        background-color: #b30000;
    }

    /* Card */

    #card{
        width: 350px;
        margin: 30px auto;
        padding: 20px;
        background-color: white;
        border-radius: 12px;
        box-shadow: 0px 4px 12px rgba(0,0,0,0.2);
        text-align: left;
    }

    /* Card Heading */

    #card h2{
        text-align: center;
        color: #e10600;
    }

    /* Card Text */

    #card p{
        font-size: 18px;
        margin: 12px 0;
    }

    /* Horizontal Line */

    hr{
        width: 60%;
        margin: 25px auto;
    }
```

7. CODE EXPLANATION

Component	Description
<input id="teamInput">	Text field for entering team name.
<select id="teamSelect">	Dropdown menu for selecting a team.
onchange="clearInput()"	Event handler to clear text input when selecting from dropdown.
onclick="showTeam()"	Event handler to display team details when button is clicked.
document.getElementById("card").innerHTML	Updates the content of the results card dynamically.
function clearInput()	Clears the text input field.
function showTeam()	Fetches input/selection, normalizes data, and injects team stats into the card div.

8. TEST CASES AND EXECUTION DATA

Test Case	Test Scenario	Input / Action	Expected Result	Actual Result	Status
TC-01	Page Load	Open HTML file	Page loads with elements	Page loaded correctly	Pass
TC-02	Valid Team (Input)	Enter "McLaren" and click button	Details for McLaren appear	Details displayed	Pass
TC-03	Valid Team (Dropdown)	Select "Mercedes" and click button	Details for Mercedes appear	Details displayed	Pass

TC-04	Clear Input	Select team from dropdown	Text input clears	Input cleared	Pass
TC-05	Invalid Team	Enter "Test" and click button	"Team Not Found" message appears	Message displayed	Pass

9. OUTPUT

Formula 1 Team Explorer 2026

Enter or Select an F1 Team:

OR

Show Details

Team Name

Drivers:

Engine:

Position:

Wins:

Points:

Formula 1 Team Explorer 2026

Enter or Select an F1 Team:

OR

Show Details

McLaren

Drivers: Lando Norris, Oscar Piastri

Engine: Mercedes

Position: 3rd

Wins: 1

Points: 220 points

Formula 1 Team Explorer 2026

Enter or Select an F1 Team:

OR

Mercedes

▼

Show Details

Mercedes

Drivers: George Russell, Kimi Antonelli
Engine: Mercedes
Position: 1st
Wins: 8
Points: 379 points

10. RESULT

Thus, a **dynamic To-Do List webpage** was successfully developed using **HTML, CSS, and JavaScript**. The application allows users to dynamically add tasks, mark tasks as completed, and delete tasks. JavaScript DOM manipulation and functions were successfully implemented, and the application was verified using multiple test cases.